

DenseAttentionSeg: Segment hands from interacted objects using depth input

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ARTICLE INFO

Article history:

Received 30 May 2019

Received in revised form 23 March 2020

Accepted 4 April 2020

Available online 22 April 2020

Keywords:

Hand-object interaction

Semantic segmentation

Artificial neural networks

Human-computer interaction

ABSTRACT

Hand segmentation is an important task in computer vision, which is usually the foundation of hand pose recognition, hand tracking, and reconstruction. For hand segmentation, it is more challenging when the hand is interacting with objects, but handling interacting motions is more important for applications like HCI and VR. In this paper, we propose a real-time DNN-based technique to segment hand and object in interacting motions from a single depth input. Our model is called DenseAttentionSeg, which contains a dense attention mechanism which effectively fuses information in different scales and improves the quality of result with skip-connections. Besides, we introduce a contour loss in model training, which helps to generate accurate hand and object boundaries. Finally, we propose our InterSegHands dataset, a fine-scale hand segmentation dataset containing about 52k depth maps of hand-object interactions, with the ground truth segmentation masks. Our experiments evaluate the effectiveness of our techniques and datasets, and indicate that our method outperforms the current state-of-the-art deep segmentation methods in handling hand-object interactions.

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1. Introduction

Hand is an important part of our body that makes humans be able to interact with the real world and other human beings. Many studies, such as pose recognition, 3D reconstruction, human-computer interaction, AR/VR, etc., have paid lots of attention to human hands. In almost all of these hand-related tasks, it is important to first detect and segment the hands from their interacting objects and backgrounds, so hand-object segmentation is a foundation for many high-level tasks.

Many existing works consider RGB images as input to perform hand segmentation. However, RGB images can be affected easily by the color of the skin, the interacted objects and the illumination condition, which add inherent difficulties to hand-related applications. In this paper, we consider using depth images which are more robust to these factors and are also easy to obtain nowadays, as more and more consumer depth sensors are available, even on mobile devices. So we propose a real-time DNN-based hand segmentation technique that focuses on handling the challenging hand-object interaction motions.

We also propose a depth-based hand-object segmentation dataset to promote the researches on this topic. Existing depth-based hand segmentation datasets either lack object interactions [1] or are with low-resolution hands and indistinct boundaries mostly [2]. A large dataset with high-resolution hand-object interactions and delicate pixel-level labels is eagerly required to meet the demand for network training in deep learning. So we propose InterSegHands to match the requirement and accompany an auto-annotation method which is used to construct the current dataset and can be used to enlarge the dataset easily in the future. We also release the dataset¹ to promote researches in this area.

Many former works on parsing human body or body parts are based on RDF (Random Decision Forest) for real-time performance [1,3,4]. With the development of GPU devices and the popularization of deep learning, higher accuracy can be achieved with real-time performance using deep learning models. In this work, we propose DenseAttentionSeg, which contains a dense attention structure to fuse the information of different scales and guide the skip-connections. Attention mechanism has been proven to be effective in CV and NLP, which often computes the attention weights for an encoder in seq-to-seq learning [5] or for different regions of a feature map [6]. Recently, similar ideas are used in general semantic segmentation methods [7,8], but

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¹ <https://drive.google.com/file/d/1PmgR6AH6hQtY305vgYnHfEQsX41r-rYc/view?usp=sharing>.

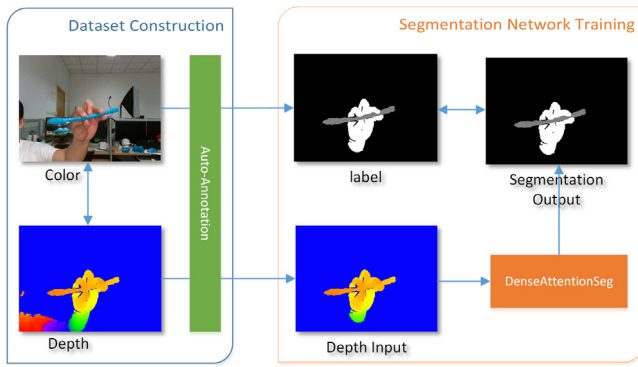


Fig. 1. The workflow of our work. The left blue container indicates our auto-annotation of dataset construction from aligned color and depth maps. The right orange container shows our DenseAttentionSeg model to segment hand from its interacting objects and generates their labels.

they are limited to high-level guidance. Nevertheless, we build an attention mechanism on different feature scales, which not only conveys coarser semantics but also finer shapes and edges information, so we call it dense attention. In addition, as hand and object boundaries are important for interacting motions, we propose a new contour loss which emphasizes the object contours and improves the final results a lot. The loss is particularly designed for this fine-scale hand-object segmentation job and will not decrease the inference speed.

The whole pipeline of our method is shown in Fig. 1. Our contributions can be summarized as follows. **First**, we introduce a large fine-scale dataset for depth-based hand segmentation in interacting motions to support relevant hand tasks, together with an auto-annotation method which can enlarge the dataset scale easily when needed. **Second**, we propose a dense attention mechanism based on an encoder-decoder network for hand segmentation, which fuses features in different scales to guide the skip-connection between the encoder and decoder. Our technique outperforms other state-of-the-art models in handling hand-object interactions. **Third**, we present a contour refinement strategy to emphasize the loss of boundary regions and increase the segmentation accuracy, which is important for small and occluded objects when handling hand-object interactions.

2. Related works

As a basic task in hand-related researches in computer vision, hand segmentation contributes to many applications, such as hand pose estimation [1,3,9,10], gesture recognition [11–13] and so on. Following, we summarize related techniques from different perspectives.

RGB-based method: There are some traditional pixel-level hand detection and segmentation methods based on RGB camera, which can be classified into 3 categories [14]: (1) local appearance-based detection, which uses local color of hand regions as a simple yet strong feature for detecting skin-color regions [15,16]; (2) global appearance-based detection, which adopts a global hand template as the search space [17,18]; (3) motion-based detection, which takes the ego-motion of the camera into account [19,20]. The deep learning methods are becoming much more powerful nowadays. [21] segmented hands egocentrically in many activities via the combination of traditional and neural methods, which requires detecting the hand region first. They used a deep model to detect hand regions and then adopted GrabCut [22] to segment hands in them. [23] explored egocentric hand segmentation in social activities. They

found that simply fine-tune the RefineNet [24] model can improve the results of [21] dramatically. This showed that general segmentation deep models can also be applied to the segmentation of hands in RGB images. But all these RGB-based methods are easy to affect by the illumination conditions and the albedo of hands and objects, which may require much bigger training space or limit the application scenario.

Depth-based method: There are also some works (including hand tracking and gesture recognition) based on depth camera [13,25–27]. As small parts of their approaches, they used simple ways to detect hands, which worked for situations with no interacting objects. However, hand segmentation itself has not been fully explored yet. Some traditional methods used RDF [1,3,4]. [28] adopted a simple segmentation method by using a black wristband in their hand tracking work. They also studied the problem of hand segmentation recently [2], where they collected a hand segmentation dataset in depth consisting of about 30k samples. But their methods only generate coarse-scale segmentation results or even like a detection task. Depth-based methods are robust to illumination conditions and skin colors, which makes it more reliable in real applications.

Hand object interaction: On account of the functionality of our hands, the most common situation in our daily life is to interact with all kinds of objects. [29,30] made a threshold of skin color in HSV space to segment hands in his hand reconstruction and pose estimation works. [31,32] focused on hand tracking under occlusion situations (typically object interaction). But they used an end-to-end network to track the hands directly, instead of proposing an explicit method to segment hand and object, because of which they could only handle simple occlusion situations. [2] proposed a two-stage random decision forest to segment hand and object based on depth. Their solution runs fast while with fair accuracy. And most of their dataset samples lack close-up information of hand and object details, which makes it impossible to apply to some reconstruction works. [33] developed a method to segment different handheld objects. But their method needs manual initialization and does not segment hands. Another relevant hand-object interaction work is [23], in which they used a high-performance network to segment hand. But they adopted RGB images as input, thus has the shortages discussed above. Hand object interaction is more difficult yet more important for hand tasks, thus it is attracting more and more attention from many researchers.

Datasets: [21] introduced EgoHands for hand tasks including segmentation based on RGB images, containing 4.8k frames in total. [23] extended it for broader usage in the wild, but it is also small in quantity with no more than 2k frames in each of their sub-datasets. [34] recently proposed a larger dataset for color images, which has about 44k samples. But it is synthetic thus difficult to apply to the real world. These datasets are somewhat not enough for RGB space to generalize well beyond their training sets.

As for the depth-based dataset, [1] released NYU Hand Pose Dataset, which contains about 6k depth frames but lacks object interaction. [35] and [36] proposed a method to generate hand segmentation mask automatically with no object interaction. These automatic approaches extend the dataset size but require the user to wear colored gloves which may change the surface structure of real hands. [2] collected a total number of 27k depth images for segmentation, but most of their dataset samples do not show the fine structure of hands and lost the object information to some degree. So there is still a need for a large and fine-scale depth-based hand segmentation dataset, which can be used in more precise applications such as reconstruction and human-computer interaction.

Network-based semantic segmentation: Recently, deep neural networks are popular in many computer vision problems,

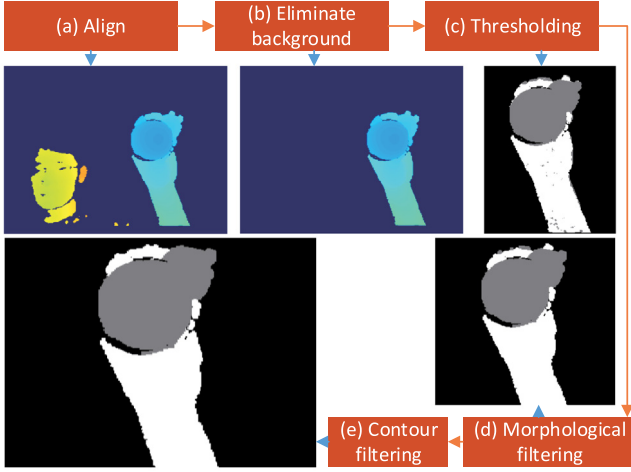


Fig. 2. The procedure of auto-annotation for building the ground truth of depth image segmentation. (a) Align the color and depth map. (b) Eliminate background using depth map. (c) Segment hand and object by HSV thresholding. (d) Morphological filtering to repair noise. See the noise dots on the wrist. (e) Contour filtering to fix system error. See the band on top of the fingers.

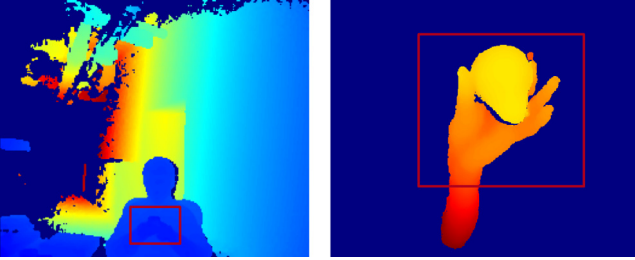


Fig. 3. Left: input sample from a recent relevant work [2]. Right: input sample from ours. The regions of interest are shown in red boxes.

such as image classification [37,38], super-resolution [39,40], retrieval [41,42] and so on. And it has been approved to solve semantic segmentation problems successfully. [43] first proposed fully convolutional neural networks. [44] designed an encoder-decoder architecture that was widely used in the following years. Other methods include Segnet [45], Fully convolutional densenets [46], PSPNet [47], RefineNet [24], Large Kernel Matters [48], MD-CCNet [49] and so on. [50–53] developed a series of Deeplab networks. There are also some methods adopt attention/supervised/fusion idea recently [7,8,54], but they are limited to high to low-level feature guidance.

Though existing network-based models work well, they were built for general segmentation tasks while may not act the same for hand-object interaction segmentation task where the classes are limited and the hand and object are closely related.

3. InterSegHands dataset

Our database contains depth images of various hand-object interacting motions and the corresponding hand masks on the depth images. To obtain the ground truth segmentation masks, we use the aligned color images. To be specific, we segment objects with predefined colors from hands on color images by a simple threshold technique performed in HSV color space. Then we transfer the masks to the depth images as depth and color images are pre-aligned.

3.1. Capture

We collect depth images of hands using Intel RealSense SR300. We keep the depth image size at 640×480 . We record images under a fixed illumination intensity and fixed color temperature for a better segmentation on color images.

3.2. Post process

Fig. 2 shows the procedure of our auto-annotation method for building the ground truth of depth image segmentation.

Background Elimination. In this step, we discard the pixels of background on both the depth and color images. First, on the depth image, we discard the pixels with depth values larger than 18 cm beyond the closest pixel and then the remaining pixels define the hand-object region. This operation is correct because when we record the dataset, we keep the hand and object the nearest parts to the camera and ensure that they extend no more than 18 cm on the depth axis. As the color and depth images are pre-aligned, we map the hand-object region masks from the depth to the color images. Notice that the alignment cannot be perfect due to our device limitations, and thus the cropped hand object region on the color image may contain some background pixels.

Segmentation. In this step, we will segment the hand from the object. This procedure is performed on the color image based on color space thresholding. As the color of the objects is chosen to be very different from skin colors, simple color space thresholding is able to segment skin from the colored objects. Notice that since color does not affect the recording of depth images, this operation will not influence the generality of our depth dataset. Here we convert color images into HSV space and set linear thresholds on the three channels. To reduce isolated segments after the thresholding, we adopt morphological opening and closing [55] on the obtained binary masks. Remember that there are some background pixels in the input region, which we expect to exclude in the hand-object segmentation procedure. As these pixels usually form some small and lathy segments due to sensors alignment, we propose a contour filter to eliminate these segments:

$$M_o = \bigcup \{s \in \mathcal{T}(M_i) | N(s) > \theta, \frac{(\partial N(s))^2}{N(s)} < \phi\} \quad (1)$$

where M_i and M_o are the input and output masks respectively. s is a segment in the input mask. Here we use segments to represent simply connected domains in the mask. $\mathcal{T}(M)$ denotes all the segments for the mask M . The area and perimeter of a segment are represented by $N(s)$ and $\partial N(s)$. In our implementation, $N(s)$ is measured by the number of pixels in s while $\partial N(s)$ is the number of boundary pixels. θ and ϕ are the two thresholds. The obtained hand and object masks are then warped back to depth images as we need.

Finally, we manually check the segmentation results for each frame and discard frames with visible errors, which is greatly faster than pixel-wise manual annotation. Based on the statistics, the discarded frames count about 11% in total. This means over 88% of our auto-annotation results are considered to be acceptable.

3.3. Statistics

We collect a total of 51,912 pairs of depth maps and the corresponding ground truth masks. The dataset covers 28 different interacting objects, including 10 real objects, 8 plasticine-made objects, 9 painted objects and one pure hand without objects. The whole dataset is split into 46,720 pairs for training and 5,192 for

Table 1
Statistics of several hand segmentation datasets.

Dataset	Type	Annotation	# Frames	Hand resolution	Object interaction
EYTH [23]	Color	Manual	1,290	Uncertain	Partial
RHD [34]	Color	Synthetic	43,986	Middle	No
NYU [1]	depth	Automatic	6,736	Middle	No
HandSeg [35]	Depth	Automatic	265,000	Middle	No
HOI [2]	Depth	–	27,525	Low	Yes
InterSegHands (Ours)	Depth	Automatic	51,912	High	Yes

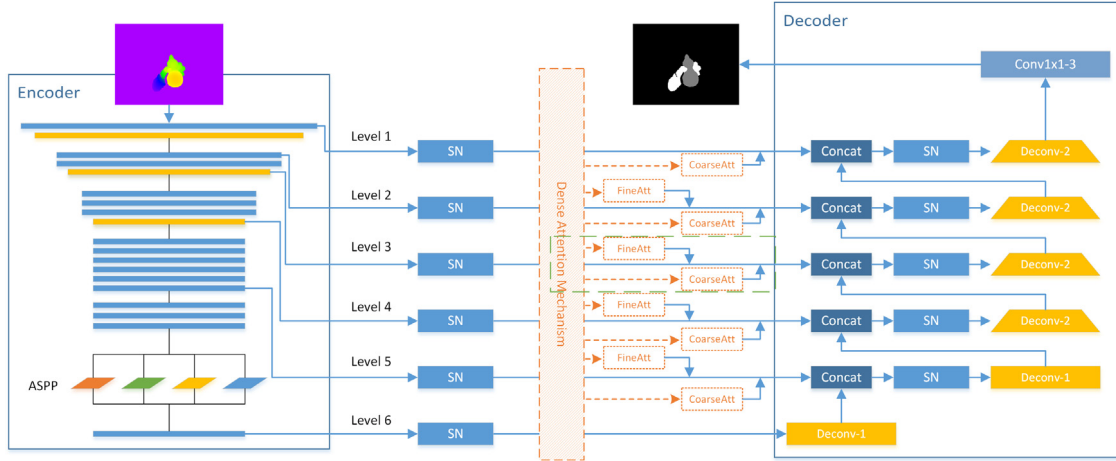


Fig. 4. DenseAttentionSeg model structure. The encoder is based on Deeplabv3 with Resnet-50 as backbone. The feature maps of the input depth image are extracted and split into different levels. The Dense Attention Mechanism guides each level of feature maps with Fine and Coarse Attention, which is applied to the skip-connections in this figure. A decoder is applied to reconstruct the segmentation labels step by step. A detailed structure of the green dashed box is shown in Fig. 5. **SN**: SqueezeNet, which is to squeeze channels of the input feature map. **Deconv**: deconvolution with stride 1 or 2. **Conv1x1-3**: final 1×1 convolution layer with softmax activation.

validation. We collect the data in a third-person perspective and Table 1 shows the statistics of our dataset as well as some other hand segmentation datasets. Fig. 3 shows our high-resolution hand area compared to the recent work [2].

4. DenseAttentionSeg

As shown in Fig. 4, we propose DenseAttentionSeg to achieve real-time hand-object segmentation from depth input, which is based on the widely used encoder–decoder network structure. In general, for encoder, we use the same structure of DeepLabv3+ [53], which generates excellent result on PASCAL VOC2012 segmentation challenge [56]. From our experiments, we find it also works well in our task. Meanwhile, we extend the decoder in DeepLabv3+. We use a densely connected attention structure to fuse all scales of features learned by the encoder and reconstruct the segmentation masks step by step. Except for the last softmax layer, all the convolution layers are followed by a batch-normalization layer [57] and a Relu activation layer [58] in our network.

4.1. Encoder

The encoder is based on DeepLabv3+ with stride-16 Resnet-50 [59] as the network backbone, together with Atrous Spatial Pyramid Pooling (ASPP). We do not use more powerful Resnet-101 and Xception as expressed in [53] for the consideration of real-time performance.

Following, we introduce the concept of levels in our encoder, which will be used in our dense attention mechanism. As the network goes deeper, the stride of feature maps gets bigger while the size gets smaller. We choose some intermediate feature maps

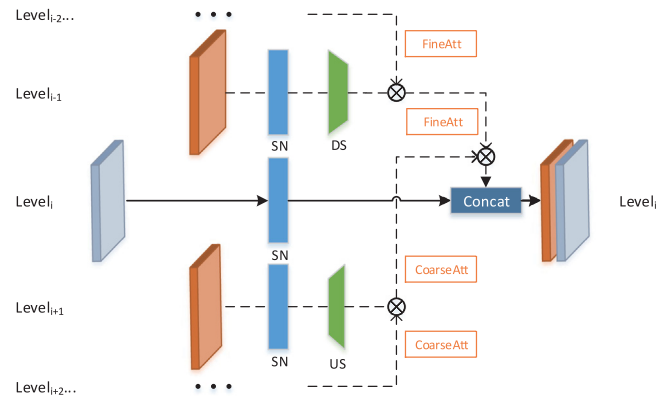


Fig. 5. Dense Attention Mechanism of level i . The FineAtt is gathered from feature maps of levels $i-1, i-2, \dots, 1$ via element-wise multiplication. The CoarseAtt is gathered from feature maps of levels $i+1, i+2, \dots, n$. **SN**: SqueezeNet, which is to squeeze channels of the input feature map. **DS**, **US**: Bilinear down-sampling and up-sampling layer respectively.

denoted as levels, which start from level 1 to level n . Specifically, we select 5 intermediate feature maps in DeepLabv3, together with the final output of ASPP (thus $n = 6$). Notice that not every level is smaller than its previous one in size, neither do Deeplabv3.

4.2. Dense attention mechanism

We propose an attention mechanism to help each level to change its features according to the information of different scales. Fig. 5 shows this mechanism in detail. We call it “dense”

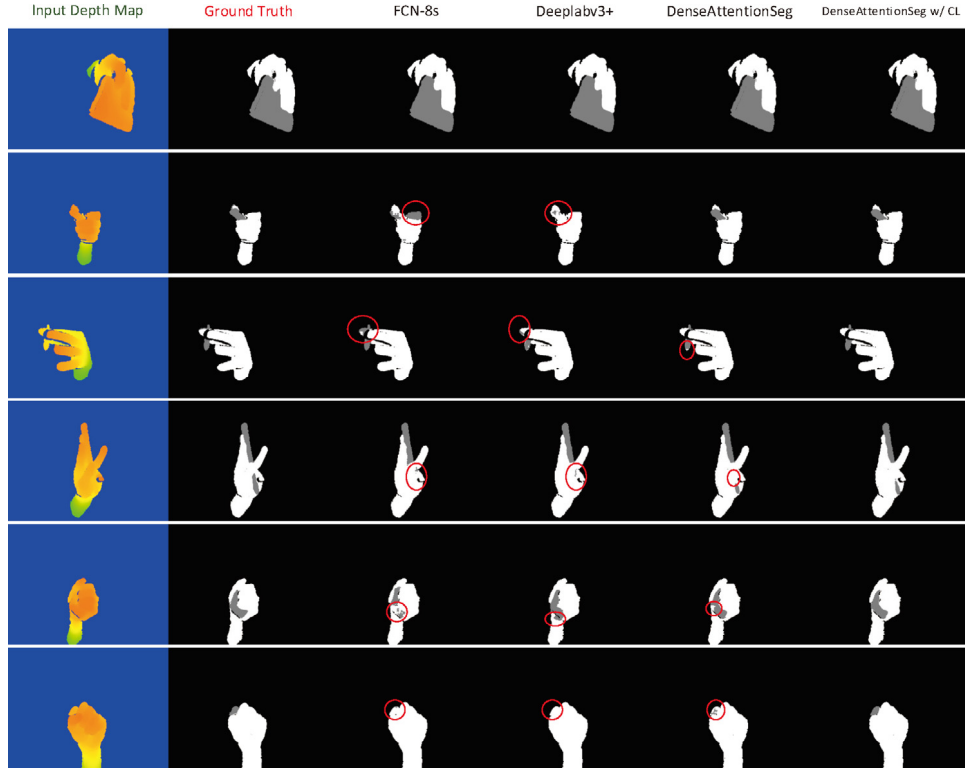


Fig. 6. Qualitative results on InterSegHands validation set. The depth map is visualized in false color. All the considered models work similarly well in easy situations (1st row), but our DenseAttentionSeg works better in more confusing (4th and 6th rows) and detailed (2nd, 3rd and 5th rows) regions, especially after contour loss finetuning (3rd, 5th and 6th rows).

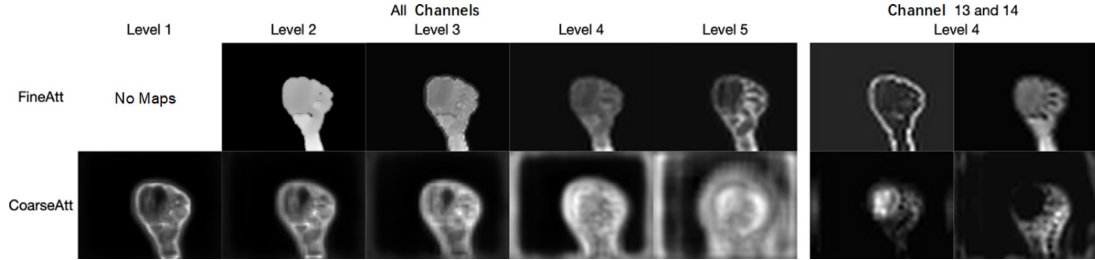


Fig. 7. Attention maps for an instance of holding a green pepper. We sum up all the feature channels together for every levels and show them as All channels on the left. We can see that the FineAtt focuses on details, while the CoarseAtt is more general. We also show two specific channels on the right where the coarse attention carries some semantic information, while the fine attention focuses on edge and shape as expected.

because each attention of every level is fused from all the other levels.

To be more specific, assuming that we are manipulating level i , the lower levels of $i - 1, i - 2, \dots, 1$ are gathered through element-wise multiplication step by step, and are used to build the *fine attention*, which conveys the shape and edge feature information to level i . Since lower levels have different numbers of feature channels and larger sizes than level i , all the lower levels are squeezed by a SqueezeNet and down-sampled by a bilinear down-sampling layer before the element-wise multiplication. Similarly, bilinear up-sampling is applied to level $i + 1, i + 2, \dots, n$ and then the *coarse attention* is constructed, which is considered to convey semantic feature information.

The SqueezeNet can squeeze the channels of the input feature map to k , which not only reduces computation costs but also unifies channels for different levels. A SqueezeNet contains a 1×1 and a 3×3 convolution layer, both with output channels of k .

4.3. Decoder

The decoder is applied to reform final segmentation labels step by step. We adopt as many as 6 deconvolution [60] layers, which guarantee that we segment hands in a close, fine-scale operating image size of 640×480 . Proved by experiments, this structure is to have higher accuracy than those with fewer decoder layers on our dataset, namely Deeplabv3+.

4.4. Loss function

We adopt **softmax cross entropy loss** as our basic training loss, as most of the other segmentation works do.

$$\mathcal{L}_{softmax} = - \sum_i y_i \log \frac{e^{x_i}}{\sum_j e^{x_j}} \quad (2)$$

In addition, we adopt a **contour loss** to refine the edges of hands and objects, which is very important for our fine-scale

segmentation task. First, we extract edges of both the network output and ground truth labels using Sobel operator, which is accomplished via convolution layers. The Sobel operation is defined as follows:

$$S(M) = \sum_{l=1}^{c-1} \sqrt{(K_h * M_l)^2 + (K_v * M_l)^2} \quad (3)$$

Where $*$ indicates convolution, M is the input map with c channels representing different classes, and M_l is the l th channel of M indicating class l . We ignore background class (with label 0) because it generates inconspicuous contours. K_h and K_v are the horizontal and vertical kernels of the Sobel operator respectively. Then, we have a contour map for both the network output and the ground truth. Next, to expand the influence of hand and object's edges smoothly, we blur the two contour maps using a Gaussian kernel:

$$B(M) = G * M \quad (4)$$

where G is the Gaussian kernel and we use a 5×5 Gaussian kernel with $\sigma = 2.121$ here. The final contour loss is defined as an L2 loss below:

$$\mathcal{L}_{contour} = \frac{1}{N} \sum (\mathcal{B}(S(M_{labels})) - \mathcal{B}(S(M_{logits})))^2 \quad (5)$$

where M_{labels} is the ground truth labels and M_{logits} is the output of our network. The equation sums up all pixels in all label classes and N is the normalization parameter. The combined loss is computed as follows:

$$\mathcal{L}_{finetuning} = \alpha \mathcal{L}_{softmax} + \beta \mathcal{L}_{contour} \quad (6)$$

where α and β are the combination weights. Notice that we use it as a fine-tuning loss after finishing the training of $\mathcal{L}_{softmax}$, because the contour is not obvious at first.

4.5. Training details

During training, we apply data augmentation by random flipping and rotation. We also add Gaussian noise to the input depth maps. The network is trained and evaluated using Tensorflow framework [61], with a batch size of 4. We use Adam optimizer [62] with an initial learning rate of 0.001, which decays by a factor of 0.8 every 80k steps to limit the highest value. The training is almost done after 100 epochs but we stop at 140 epochs to achieve every tiny improvement. Another 10 epochs are added to the fine-tuning procedure using the contour loss.

We set a high softmax loss weight of 5 for hand and object while keeping that of the background as 1. The batch-norm loss weight is 0.001, and α , β in contour loss are 1.0 and 0.005. The output channel k in SqueezeNet is set to 64.

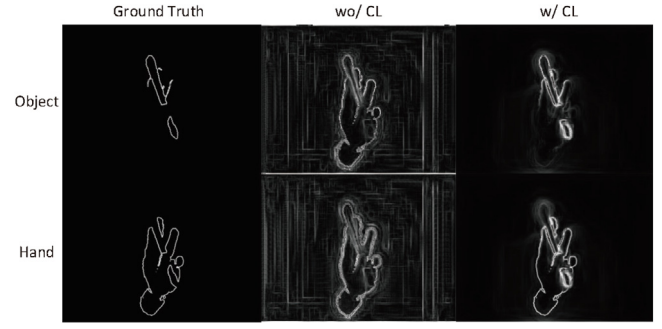


Fig. 8. Contours extracted for the object and hand. From left to right, the contours are extracted from the ground truth segmentation mask, the results without and with the contour loss. We can see that, with our contour loss finetuning, the boundaries of the results are more similar to the ground truth.

5. Experiments

In this section, we first introduce the evaluation metrics. Then we quantitatively and qualitatively compare our method with state-of-the-art models. Following, we evaluate our contour loss and dense attention mechanism. After demonstrating the generalization and applications of our technique, we introduce the current limitations.

5.1. Evaluation metrics

Our method aims to segment all pixels of a depth image into three classes - hand, object and background. To evaluate the result, we use the four metrics in [43], which are pixel accuracy, mean accuracy, mean IU and frequency weighted IU. Considering that our key objective is to segment hands from objects, we also compute the mean IOU (Intersection over Union) for the hand and object classes.

5.2. Segmentation results

We train different models on the training set of our InterSegHands and evaluate them on the validation set. The result is shown in Table 2. Our DenseAttentionSeg with contour loss finetuning gains 93.94% mean IOU, which is 1.18% higher than the baseline method DeepLabv3+(Resnet-50, OS-16) [53], and also outperforms other state-of-the-art methods for universal segmentation tasks, while still keeps high inference speed.

Besides the quantitative evaluations, we also perform qualitative comparisons. In Fig. 6, we show several results in the InterSegHands validation set. In easy cases where hand and object are very different and have smooth and clear boundaries, all the compared methods work well (1st row). But for complicated

Table 2

Quantitative results on InterSegHands validation set. The **mean IOU** is per class mean intersection-over-union rate for hand and object classes. The **time** is the inference time of batch size 1 on GTX Titan Xp with an input size of 640×480 . **CL** means contour loss finetuning.

	Mean IOU	Pixel acc.	Mean acc.	Mean IU	f.w. IU	Time
FCN-32s [43]	87.60	99.28	94.12	91.69	98.6	35.4 ms
FCN-16s [43]	87.59	99.39	94.48	91.67	98.83	35.6 ms
FCN-8s [43]	91.58	99.62	96.32	94.38	99.26	33.5 ms
Large Kernel Matters (k = 9) [48]	92.55	99.69	96.95	95.10	99.40	49.5 ms
RefineNet [24]	92.53	99.64	96.74	94.98	99.30	61.9 ms
DeepLabv3+(Resnet-50, OS-16) [53]	92.76	99.64	96.99	95.16	99.31	47.8 ms
DenseAttentionSeg	93.65	99.74	97.48	95.77	99.50	53.7 ms
DenseAttentionSegw/ CL	93.94	99.75	97.67	95.99	99.51	53.7 ms

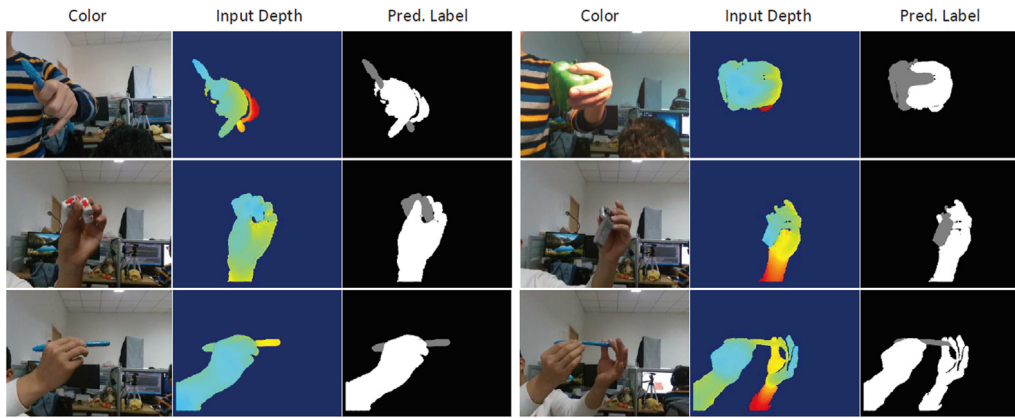


Fig. 9. Generalization analysis. **1st row:** objects that are used in the training dataset are manipulated with a different person. Notice that the case on the right is in an extreme illumination condition. **2nd row:** objects that have not appeared in the training dataset are manipulated. **3rd row:** we handle the right hand (on the left) and both hands (on the right) cases which are not contained in the dataset.

Table 3

Evaluation of dense attention mechanism which increases the value of mean IOU. Though our model is slightly slower than the baseline due to our heavier memory access cost (MAC), it is more lightweight both in computations and parameters because of our better decoder design.

	Mean IOU	#flops	#params	Time
Deeplabv3+ (baseline)	92.76	168.7G	40.34M	47.8 ms
Ours wo/attention	92.81	145.2G	39.91M	51.1 ms
Ours w/attention	93.65	146.0G	39.93M	53.7 ms

cases, different methods generate different artifacts. For example, the light-weight FCN-8s fails to distinguish hand and object regions in some examples (2nd, 4th and 6th rows) and cannot identify clear boundaries in some local regions (3rd and 5th rows). Deeplabv3+ is somewhat better but still generates noticeable errors (2nd, 4th, 5th and 6th rows). Our DenseAttentionSeg improves the results a lot (2nd, 3rd and 4th rows). Even for small and heavily occluded objects (2nd, 3rd and 4th rows), our model is still robust and generates promising results, which are more accurate on edges after the contour loss finetuning (2nd and 4th rows)

5.3. Evaluation of dense attention mechanism

Our dense attention mechanism aims to guide the skip-connections between the encoder and decoder to merge high-level semantic and low-level edge information together. In our experiment, the dense mechanism provides a positive effect on our model. Specifically, when we drop the dense attention mechanism, our model gains a mean IOU of 92.81%, which is still a little bit higher than our baseline model Deeplabv3+, which may be due to our better feature reforming decoder. After adding the dense attention mechanism, the mean IOU rises by 0.84%. The results and complexity analysis are shown in Table 3. We also show the attention maps in Fig. 7, in which we can see the FineAtt plays a more important role in extracting low-level features while the CoarseAtt assists more in extracting high-level features.

5.4. Evaluation of contour loss

The quantitative and qualitative comparisons of our method with and without the contour loss are also shown in Table 2 and Fig. 6 respectively. We see that the contour loss further improves the quality of our results on all the metrics. Visually, we can see that the improvements are majorly lying on local boundaries as expected. This is important because the boundaries

are significant for accurate hand reconstruction and hand-based remote manipulation. Another way to visualize the effectiveness of our contour loss is to perform boundary detection on the ground truth segmentation mask, the results without and with the contour loss as shown in Fig. 8, where the contour maps are extracted as Eq. (5) does. We can see that the finetuning guarantees more clear and accurate hand-object boundaries.

5.5. Generalization and application

We show some real-time testing samples in Fig. 9. Samples in the 1st row include objects from the training dataset manipulated by a different person. Our method distinguishes the stick and the little finger clearly in the left case. Also, depth input makes the method work in extreme illumination conditions in the right case. For the objects that are not seen in the training set, our model is still robust, as shown in the 2nd row, where even the flexible watchband can be segmented correctly as shown in the right case. In addition, both our training and validation sets contain only left-hand samples, but our model can still handle the right hand and both hands cases in the 3rd row, because of our data augmentation strategies and the fully convolutional structure in our model.

Our hand segmentation may enable many hand applications. We show one example on a 3D hand tracking task [63] in Fig. 10. Manipulating objects is a common case in hand tracking. With our hand segmentation, the tracking procedure is less affected by the object. And if needed, it will be possible to track both hands and objects with our segmentation technique.

5.6. Analysis on failure Cases

Our depth-based model is robust to hand object appearances and illuminations, but sometimes, it may be tricked by objects which are quite close to the hand or have similar shapes to hand or hand components (like figures), as shown in the first two rows of Fig. 11. Apart from that, some black objects are difficult to be recorded by current depth sensors, as shown in the 3rd row. Also, depth sensors using structured light cannot perform well in outdoor scenes.

6. Conclusion and future works

We have proposed a deep model - DenseAttentionSeg to perform hand-object segmentation of interacting motions, which is



Fig. 10. Applying our method for a hand tracking task [63]. We show the tracking hands on the right two columns and the gray shades are the recorded depth point clouds. Our segmentation helps to improve the hand-tracking results in the situation of hand-object interaction.

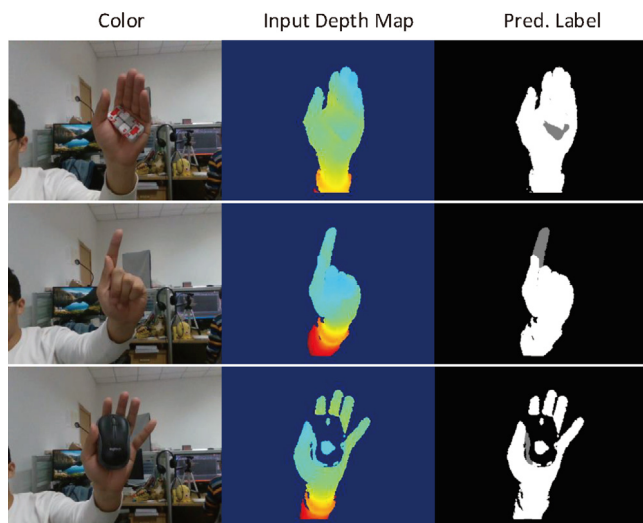


Fig. 11. Failure cases. **1st row:** it is challenging when the surface of the object is close to the hand skin. **2nd row:** some special poses may trick our model to distinguish objects and hand components. **3rd row:** if the camera fails to record depth data, our model will fail.

trained on our InterSegHands Dataset - a 52 K fine-scale hand-object segmentation dataset of depth images. Our DenseAttentionSeg contains a dense attention mechanism to fuse all *levels* features as attention to guide the skip-connection between the encoder and decoder, and a contour loss which helps to train the model to get more precise results on hand and object boundaries. According to our evaluations, our DenseAttentionSeg achieves state-of-the-art performance among several advanced models on our task and is applicable for hand tracking to handle complicated cases as the good generalization capability of our technique.

Besides hand tracking, our segmentation may be applied in other hand-related applications in the future, such as pose recognition, hand and object reconstruction and so on. We also notice that we may combine both depth and color maps to overcome the limitations of using a single depth or RGB sensor, but it might lead to more advanced dataset acquiring strategy and models with higher computing power, which will be worthy to explore in the future.

CRediT authorship contribution statement

Zi-Hao Bo: Conceptualization, Methodology, Software, Data curation, Validation. **Hao Zhang:** Writing - original draft, Validation. **Jun-Hai Yong:** Project administration. **Hao Gao:** Supervision. **Feng Xu:** Conceptualization, Funding acquisition, Supervision.

Acknowledgments

This work was supported by the National Key R&D Program of China 2018YFA0704000, the NSFC (No. 61822111, 61727808, 61671268) and Beijing Natural Science Foundation (JQ19015, L182052).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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